Dave Dalcin

Student ID: 93315

Big Data and Data Science

Swiss School of Business and Management Geneva

From Clickstream Data to Purchase Prediction: A Practical Decision Tree Application

Course Lecturer: Sabrina Šuman, PhD, Senior Lecturer

Table of Contents

1. Brief Description of the Practical Example

Project repository:
https://github.com/davedalcin/Big-Data-and-Data-Science/tree/main/Assignment-2-Practical

Supporting Python notebook: The assignment 2 notebook in the repository contains the full analysis and should be read alongside this documentation.

E-commerce stores observe what visitors do while browsing pages, looking at products and potentially completing a purchasing transaction. This behaviour can help predict their future behaviour in an e-business. Such observed behavior is collected and needs to be used for predictive analysis as per a specific business purpose. The data has to be then prepared, analyzed and evaluated.

Based on the discussion above, the first part, in practice this section of the project constructs an interpretable classification task using the Online Shoppers Purchasing Intention Datasets (Sakar & Kastro, 2018). The 12330 sessions were recorded with behavioural, web analytics, temporal, technical and visitor type related variables gathered. The output (whether a purchase is made to generate some revenue or not), which is dichotomous, served as the ground-truth result. Thus, the model aimed to predict the behavioural outcome rather than the psychology underlying.

The chosen approach of Decision Tree due to its rules could be seen directly. The analysis of each part includes getting better understanding on the source data, the preprocessing of categorical with stratified split into training and test, majority class modelling, the cross-validated selection of models and the evaluation measures in terms of accuracy, balanced accuracy, precision/recall, F1-score and confusion matrix. Measures focused on the purchase class are needed, as purchasing sessions comprise a minority of observations.

The model structure and feature importance are also examined. As a sensitivity analysis, since PageValues turned out to be by far the leading feature, PageValues is excluded from the predictors and the entire model selection process is rerun to verify if other variables have values predicting sales - and to answer the question, at the time of the customer's visit perhaps this PageValues data is not available at hand for prediction.

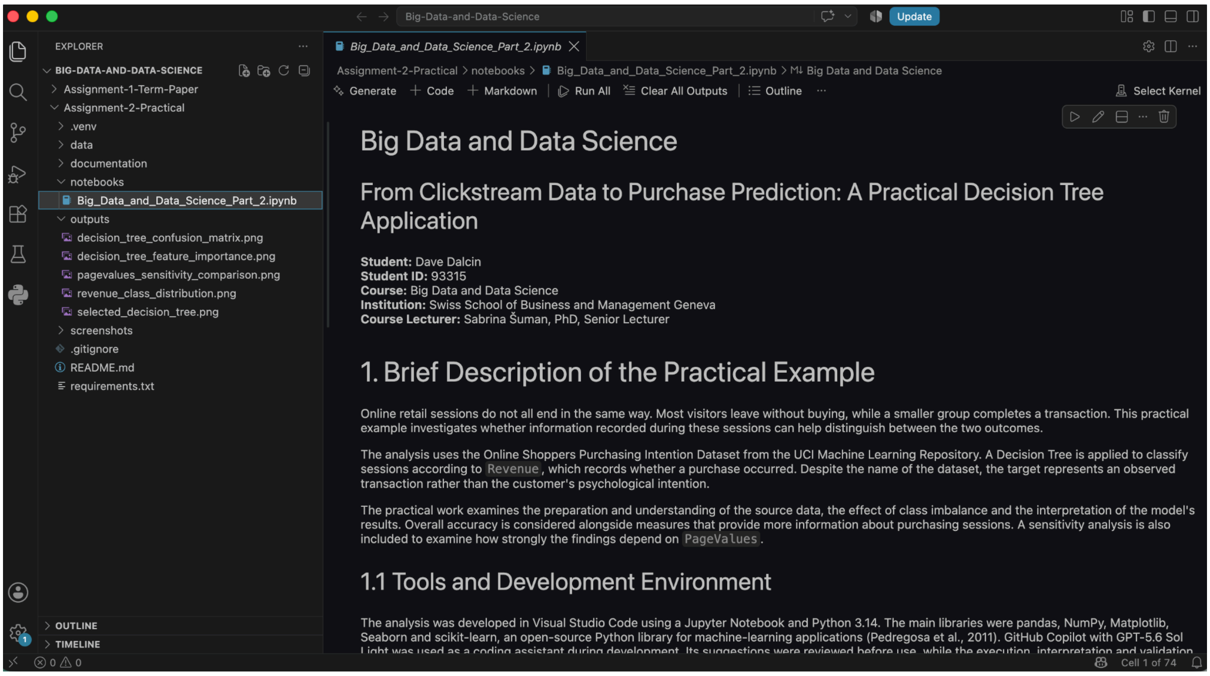
The example demonstrates the relationship of the statistical performance with possible usability for the business. Predictions could be used for recommendations, service prompts, selective incentives, but it is not possible to claim to prove intervention leading purchase. The usefulness also depends when and for the costs would be needed to make a decision and the cost of false-positive/false-negative decisions.

1.1 Tools and Development Environment

The practical analysis was developed in Visual Studio Code in a Jupyter Notebook using Python 3.14. Data preparation and initial analysis were undertaken in pandas and NumPy. Graphs and figures were produced via Matplotlib and Seaborn, and the preprocessing, model selection, and model evaluation steps were taken with scikit-learn (Pedregosa et al., 2011).

Where applicable, a fixed random state was used to support reproducibility. Dataset, notebook, generated figures, screenshots, documentation and dependency list are organized in different project folders. GitHub Copilot with GPT-5.6 Sol Light was used as a coding assistant in this project. Suggestions by Copilot were all reviewed prior to usage, and the responsibility for code execution, methodological choices, interpretation and validation remained with the author.

Figure 1. Practical project and development environment



Source: Author’s project environment.

Description: The practical analysis was developed in Visual Studio Code using a Jupyter Notebook. Project structure separates the source dataset, notebook, generated outputs, screenshots, documentation and dependency file to support organisation and reproducibility.

2. Source Data Preparation and Understanding

2.1 Dataset Background

The data source for the practical analysis is the Online Shoppers Purchasing Intention Dataset. Researchers Sakar and Kastro (2018) uploaded their dataset to the UCI Machine Learning Repository. These researchers study the real-time prediction of online purchasing behaviour based on online session details (Sakar et al., 2019).

This dataset comprises 12,330 observations collected over a period of one year. The developers of this dataset state that each observation represents a session belonging to a different user to ensure that observations from the same customer, campaign, special day, or period are not concentrated. That is, each observation represents one session, which was of a single user.

Each session has 17 predictors and the binary target Revenue. The predictors concern page activity, time spent on different page categories, the web analytics measures and some contextual information such as month, visitor type, traffic type and weekend access. Revenue is True when the session leads to a transaction and False otherwise.

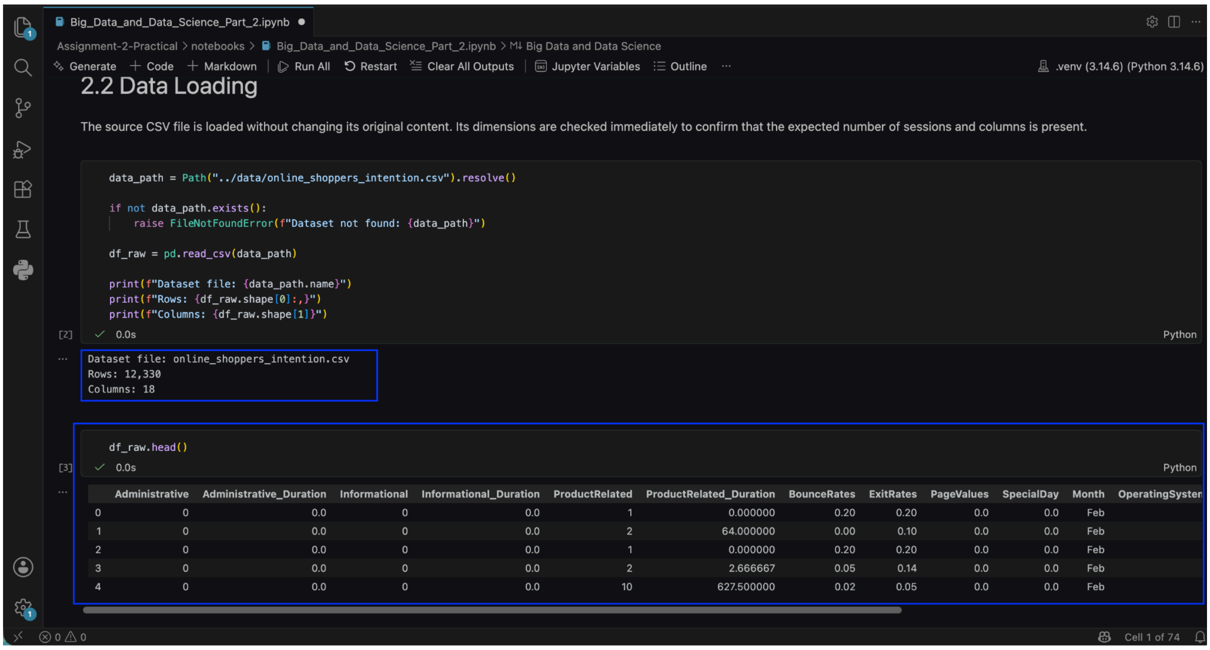
The data set can be used to illustrate classification, but it cannot be described as a complete Big Data environment. The cases consist of aggregated observations on a session rather than a stream of click events or entire customer history and activity across different devices or channels. This data set would also be enough for the purposes of teaching Data science within the context of Big Data and predictive analytics examined in the theoretical paper.

2.2 Data Loading and Initial Inspection

The original CSV file has been loaded into the pandas DataFrame as is, no source values have been altered, and the number of rows and columns has been checked immediately. As expected, there are 12,330 sessions and 18 columns, consisting of 17 predictors plus the target.

The first observations were then displayed to ensure it had been read correctly and offer a view of the values available. The preview showed numerical page counts and durations, web analytics measures, categorical codes, Boolean values, and the Revenue outcome.

Figure 2. Source dataset loading and initial preview



Source: Author's analysis based on Sakar and Kastro (2018).

Description: The output confirms that the source file was loaded successfully and contains the expected dimensions. The initial rows illustrate the combination of numerical, categorical and Boolean information available for each session.

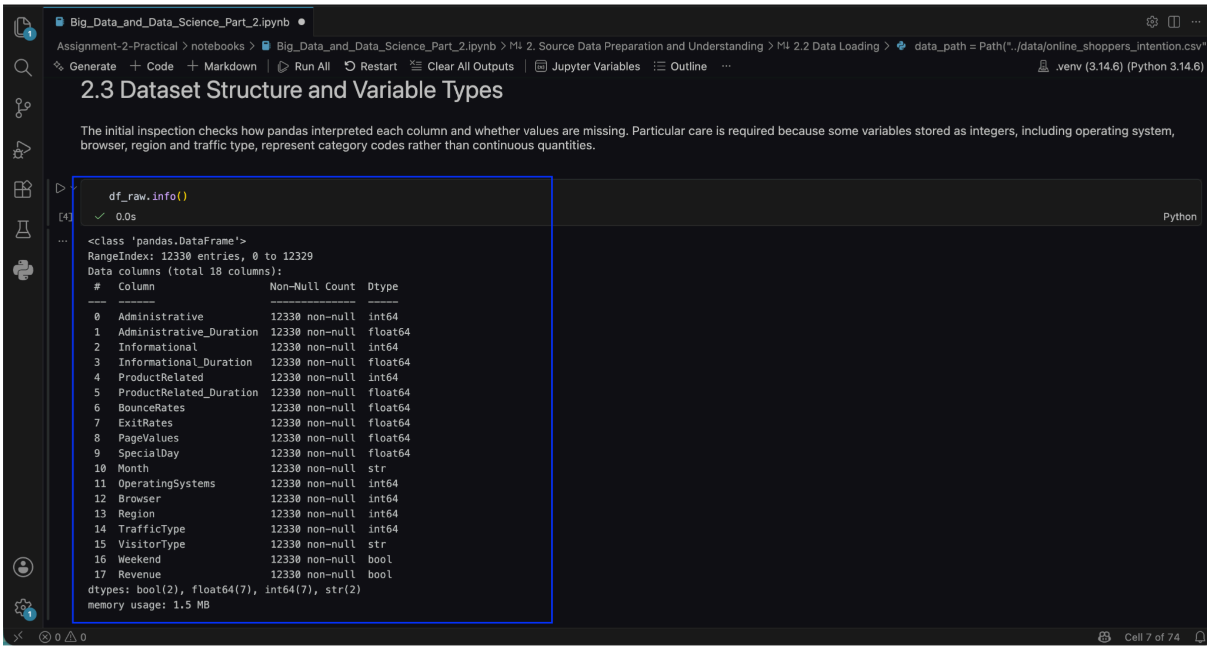
2.3 Dataset Structure and Variable Roles

The analysis of the dataset showed 7 floating-point variables, 7 integer variables, 2 text variables and 2 Boolean variables. Columns have no missing values. The treatment of each variable, however, was not decided merely by its storage type.

Administrative, Informational and ProductRelated are page counts and their respective durational variables are the time spent on those page categories. BounceRates, ExitRates, PageValues and SpecialDay are numerical. But OperatingSystems, Browser, Region and TrafficType contain integer codes for respective categories instead of any measurable quantity.

A larger browser or region code does not represent greater or more of anything; treating these codes numerically would be assigning an artificial ordering. These were treated as categorical predictors and therefore subject to one-hot encoding.

Figure 3. Dataset structure and variable types



Source: Author’s analysis.

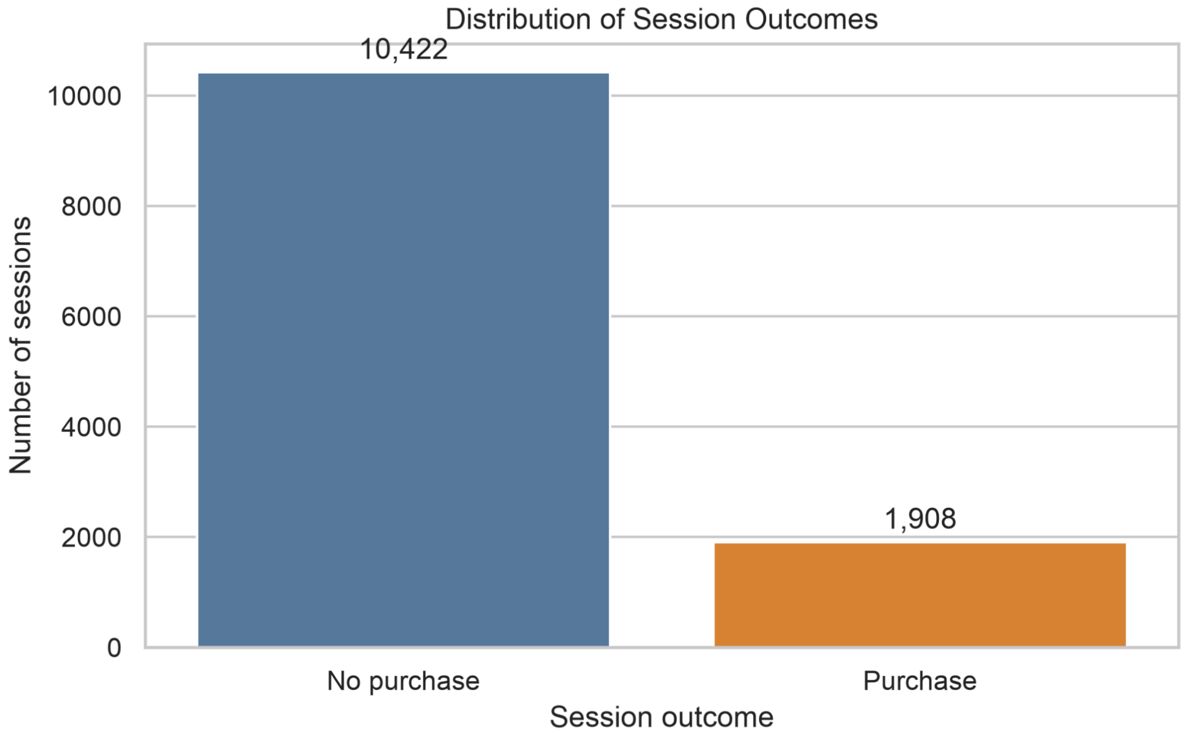
Description: The inspections verifies that all cases are complete, the storage type of each variable and the number of values assumed by it. This is helpful to be able to tell the difference between actual numerical measures and categories for which digits have been used in coding the values.

2.4 Target Distribution and Class Imbalance

The target variable is unevenly distributed. Out of 12,330 sessions, 10,422 did not result in a purchase and 1,908 sessions did. So 15.47% of sessions resulted in a purchase whilst 84.53% did not.

The implication is that a classifier could assign “No Purchase” for each observation and achieve 84.53% accuracy without being able to identify a purchasing session. Overall accuracy was used as a measure; however, this was not considered in itself enough, and balanced accuracy, precision, recall, F1-score and the confusion matrix were introduced as the other measures on account of the fact that they can provide more detail about performance of the two classes.

Figure 4. Distribution of session outcomes



Source: Author's analysis based on Sakar and Kastro (2018).

Description: Figure 4 shows that there is a severe class imbalance between purchasing and non-purchasing sessions that sets the majority-class baseline and explains the evaluation of the purchase class using purchase-class evaluation metrics.

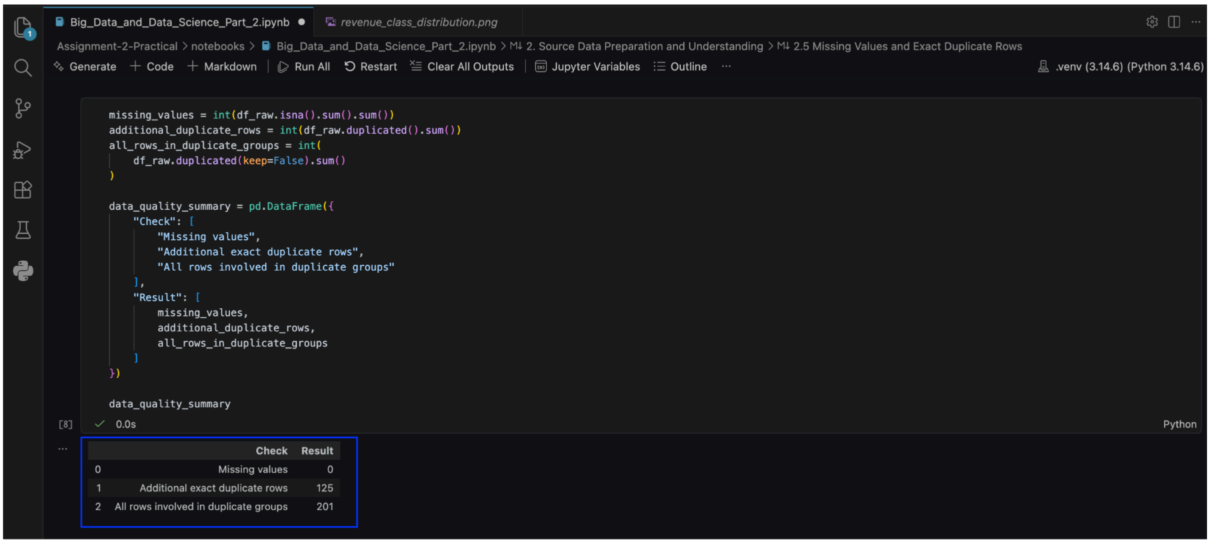
2.5 Missing Values and Exact Matching Rows

The data-quality assessment resulted in no missing data values and the duplicate checker found an extra 125 exact matches, while 201 records were present in duplicate groups. The matching records belonged to the non-purchase class only.

The identical records were retained. The dataset documentation mentions that each record represents a different user session, but the file doesn't include a session identifier that would confirm whether the matching values are caused by accidental duplication. Observations are aggregated; hence the same records may occur legitimately when generated by different sessions.

The deletion of these would remove records from the majority class, thus altering the representation of purchase prevalence; therefore, retaining the rows was justifiable. This is another area within the sample data source where a session identifier would have allowed confirmation of a problem that cannot currently be confirmed and should be listed as a limitation of the data source.

Figure 5a. Missing-value and exact-match assessment



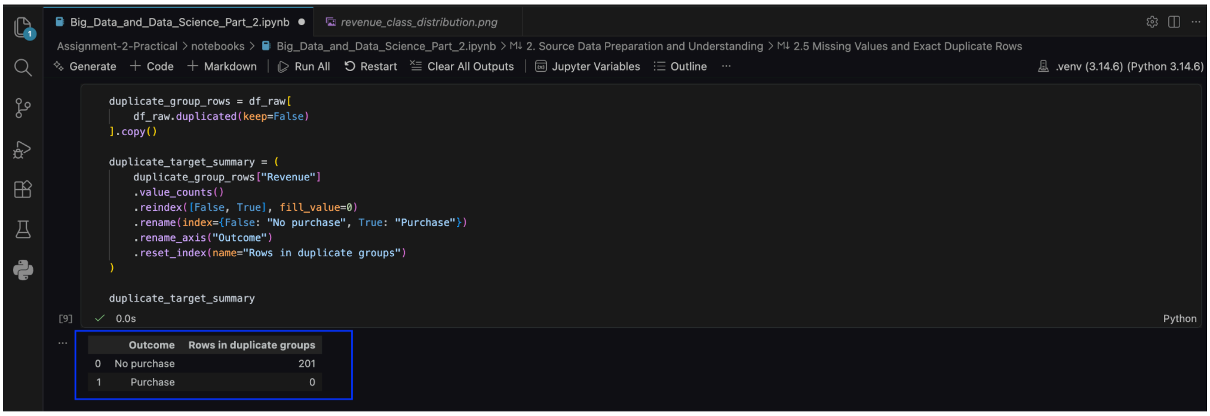
```python
missing_values = int(df_raw.isna().sum().sum())
additional_duplicate_rows = int(df_raw.duplicated().sum())
all_rows_in_duplicate_groups = int(
    df_raw.duplicated(keep=False).sum()
)

data_quality_summary = pd.DataFrame({
    "Check": [
        "Missing values",
        "Additional exact duplicate rows",
        "All rows involved in duplicate groups"
    ],
    "Result": [
        missing_values,
        additional_duplicate_rows,
        all_rows_in_duplicate_groups
    ]
})

data_quality_summary
```

	Check	Result
0	Missing values	0
1	Additional exact duplicate rows	125
2	All rows involved in duplicate groups	201

Source: Author's analysis based on Sakar and Kastro (2018).

Description: There were no missing values detected. Additionally, there were 125 additional exact-match rows after their first appearances and a total of 201 observations belonging to exact-match groups after all occurrences were considered.

Figure 5b. Distribution of exact-match rows by purchase outcome



```python
duplicate_group_rows = df_raw[
    df_raw.duplicated(keep=False)
].copy()

duplicate_target_summary = (
    duplicate_group_rows["Revenue"]
    .value_counts()
    .reindex([False, True], fill_value=0)
    .rename(index={False: "No purchase", True: "Purchase"})
    .rename_axis("Outcome")
    .reset_index(name="Rows in duplicate groups")
)

duplicate_target_summary
```

	Outcome	Rows in duplicate groups
0	No purchase	201
1	Purchase	0

Source: Author's analysis based on Sakar and Kastro (2018).

Description: All 201 observations within groups of exact matches were labelled as non-purchasing sessions. The observations were retained because it cannot be certain that recordings with identical attributes reflect erroneous duplicates, as distinct sessions may result in the same values for the existing variables.

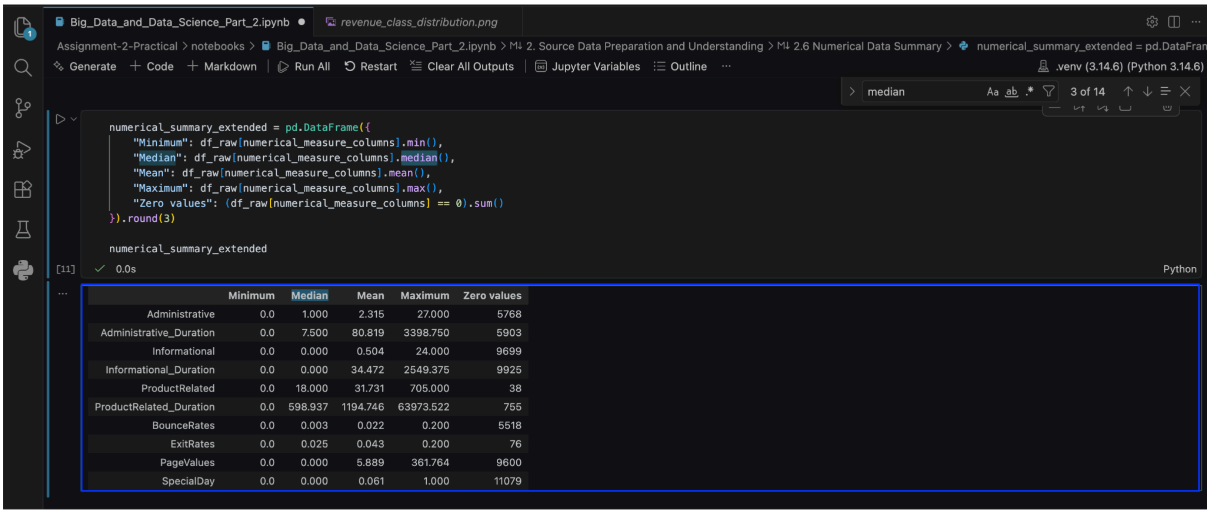
2.6 Numerical Data Summary

The numerical behavioural measures are generally right-skewed - most of the sessions involve a relatively limited amount of activity, but some sessions have a much larger number of page views or longer durations. For instance, the median value for ProductRelated_Duration is 598.937 while the mean value is 1,194.746 (and the maximum value is above 63,000).

Zero values are found frequently as well. However, a zero value cannot usually be treated as missing: if a visitor did not even visit a certain type of page during the session, it is also a zero. Informational pages were not visited in 9,699 sessions; PageValues was zero in 9,600 sessions. Most sessions also got a zero for SpecialDay because a session that is not close in time to any special business date receives a zero.

The highest values were retained (far away from the median value), as long sessions and extensive browsing could portray realistic behaviour and observations did not necessarily seem to represent errors. No feature scaling was performed; Decision Trees split numerical variables using thresholds.

Figure 6. Summary of numerical behavioural measures




```python
numerical_summary_extended = pd.DataFrame({
    "Minimum": df_raw[numerical_measure_columns].min(),
    "Median": df_raw[numerical_measure_columns].median(),
    "Mean": df_raw[numerical_measure_columns].mean(),
    "Maximum": df_raw[numerical_measure_columns].max(),
    "Zero values": (df_raw[numerical_measure_columns] == 0).sum()
}).round(3)

numerical_summary_extended
```

	Minimum	Median	Mean	Maximum	Zero values
Administrative	0.0	1.000	2.315	27.000	5768
Administrative_Duration	0.0	7.500	80.819	3398.750	5903
Informational	0.0	0.000	0.504	24.000	9699
Informational_Duration	0.0	0.000	34.472	2549.375	9925
ProductRelated	0.0	18.000	31.731	705.000	38
ProductRelated_Duration	0.0	598.937	1194.746	63973.522	755
BounceRates	0.0	0.003	0.022	0.200	5518
ExitRates	0.0	0.025	0.043	0.200	76
PageValues	0.0	0.000	5.889	361.764	9600
SpecialDay	0.0	0.000	0.061	1.000	11079

Source: Author's analysis based on Sakar and Kastro (2018).

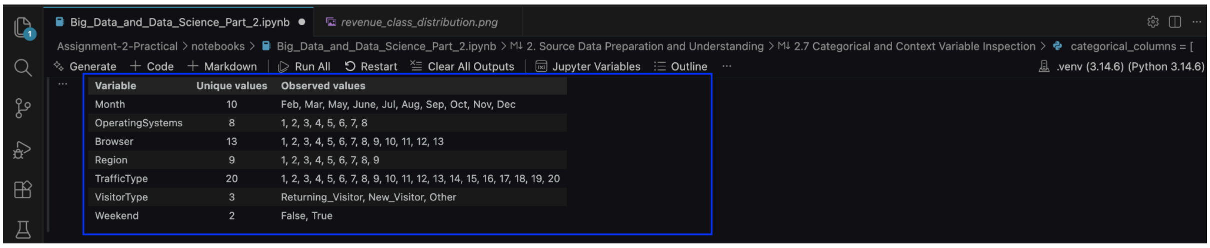
Description: Output showed substantial right-skewness and prevalence of zero values across several numerical measures. Observations were retained given zero values can reflect sessions containing no particular activity and unusually high values may indicate actual browsing behaviour. Numerical scaling was not required by the Decision Tree model.

2.7 Categorical and Context Variable Inspection

Only 10 months were found by categorical inspection. The months missing are January and April, which is a characteristic of the source file, not a missing-value problem. The months containing the highest number of sessions are May and November.

The category of Returning Visitors consists of the largest group of visitors: 10,551 sessions, compared to 1,694 new visitors and 85 sessions related to the Other group. OperatingSystems, Browser, Region and TrafficType are represented by codes, while their original meanings are not provided. These variables may be used to differentiate the categories when conducting the modelling; however, since the corresponding browsers, regions and traffic sources are unrecognisable through the codes, decisions cannot be made about a particular browser, location or traffic source based on assumptions without solid evidence.

Figure 7. Observed categorical and contextual values



Variable	Unique values	Observed values
Month	10	Feb, Mar, May, June, Jul, Aug, Sep, Oct, Nov, Dec
OperatingSystems	8	1, 2, 3, 4, 5, 6, 7, 8
Browser	13	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13
Region	9	1, 2, 3, 4, 5, 6, 7, 8, 9
TrafficType	20	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20
VisitorType	3	Returning_Visitor, New_Visitor, Other
Weekend	2	False, True

Source: Author's analysis based on Sakar and Kastro (2018).

Description: The output confirms categories observed in each of the context variables. The integer-coded technical and traffic variables are listed as nominal labels because their values do not provide an ordered scale, which made it possible to encode them as categorical variables in the preprocessing stage.

2.8 Preliminary Relationships with Revenue

Descriptive statistics were compared before modelling to find out differences between sessions that involve a purchase and those that do not. Medians are used for the first numerical comparison since several behavioural measures are strongly skewed. The relationships were treated as associations rather than evidence that one variable causes purchasing.

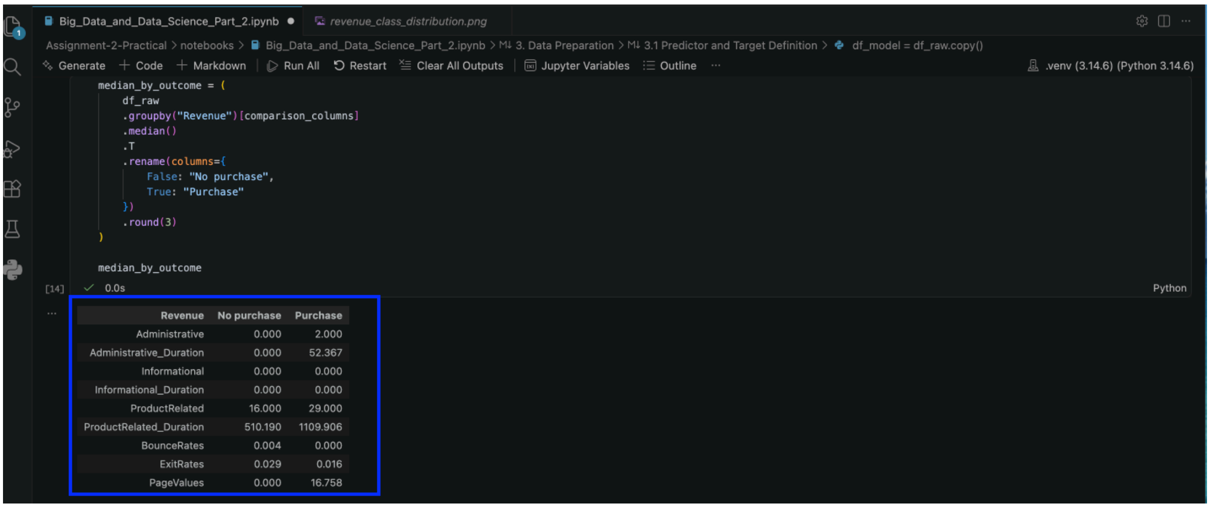
Purchasing sessions tended to involve more pages related to the product: their median in this category was 29, while non-purchasing sessions had 16. Their median duration spent on product-related pages was more than twice as much as that in sessions without a purchase. Median bounce and exit rates were higher in non-purchasing sessions, which supported that visitors did not progress as far through the website.

The most obvious difference was with PageValues. Here the median was zero for non-purchasing sessions and 16.758 for purchasing sessions. This strong separation indicated that the variable might have substantial use in classification. However, with this finding there also emerged a practical question: the usefulness of this variable as an intervention tool relies on whether it can be accessed and calculated within the required decision period or not.

New and Returning visitor groups were also explored, showing 24.91% conversion for new visitors and 13.93% for returning visitors. Further to that was an option 'Other', resulting in an 18.82% conversion but referring to just 85 sessions. Differences here should not necessarily be considered causal because visitor groups may have different purposes, arrive from different sources or show different browsing behaviour.

Purchase rates differed across different months. The month of February had the lowest purchase rate of 1.63%. The month of November had the highest purchase rate of 25.35%. This could have been affected by different seasonality factors, special campaign offers or differences in website traffic generation. Month-related indicators may provide some level of prediction. However, the results may not be sufficient to prove that actual transactions occurred due to calendar month-based influences.

Figure 8. Median session characteristics by purchase outcome



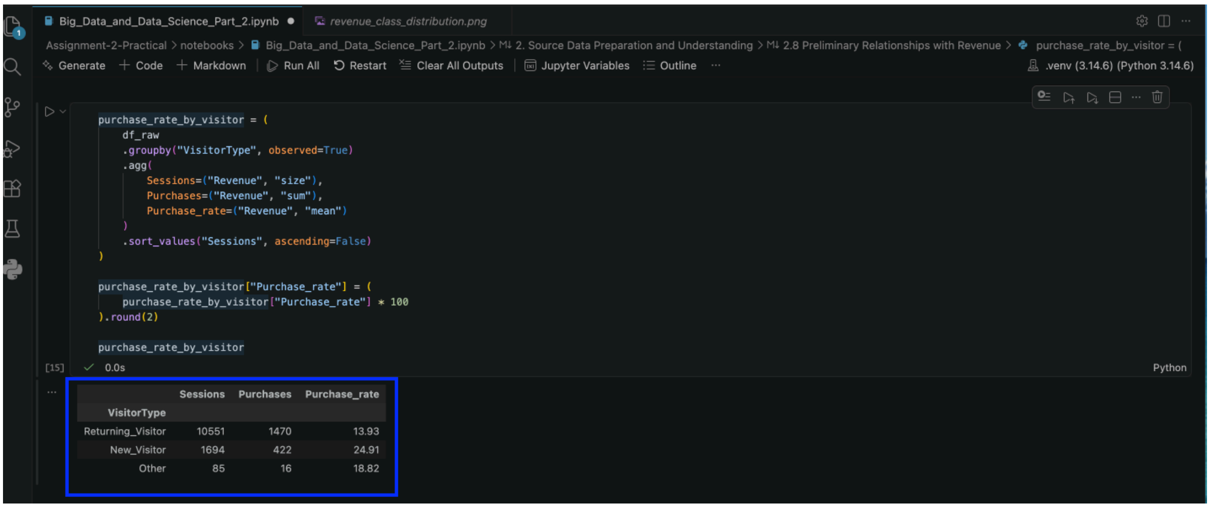

```python
median_by_outcome = (
    df_raw
    .groupby("Revenue")[comparison_columns]
    .median()
    .T
    .rename(columns={
        False: "No purchase",
        True: "Purchase"
    })
    .round(3)
)

median_by_outcome
```

Revenue	No purchase	Purchase
Administrative	0.000	2.000
Administrative_Duration	0.000	52.367
Informational	0.000	0.000
Informational_Duration	0.000	0.000
ProductRelated	16.000	29.000
ProductRelated_Duration	510.190	1109.906
BounceRates	0.004	0.000
ExitRates	0.029	0.016
PageValues	0.000	16.758

Source: Author's analysis based on Sakar and Kastro (2018).

Description: Figure 8 shows that engagement was relatively stronger during purchasing sessions compared with non-purchasing sessions, both in terms of increased product activity and reduced exit activity. PageValues showed the largest amount of separation; hence, feature importance analysis and sensitivity analysis were done targeting this variable.

Figure 9a. Purchase rates by visitor type



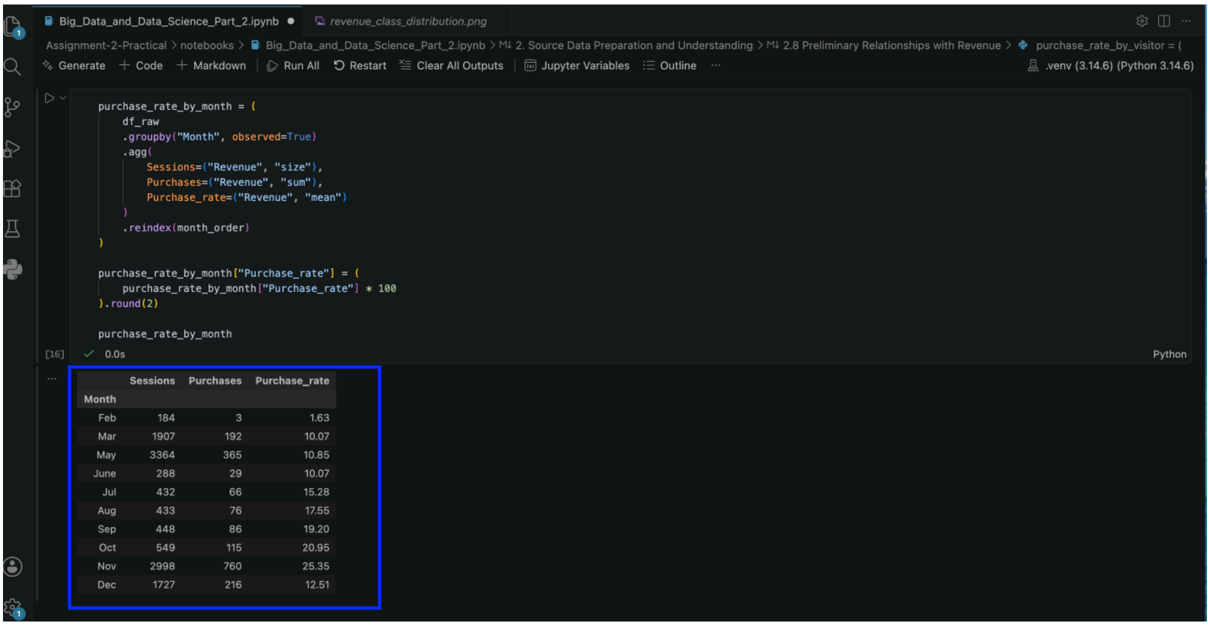
```python
purchase_rate_by_visitor = (
    df_raw
    .groupby("VisitorType", observed=True)
    .agg(
        Sessions=("Revenue", "size"),
        Purchases=("Revenue", "sum"),
        Purchase_rate=("Revenue", "mean")
    )
    .sort_values("Sessions", ascending=False)
)

purchase_rate_by_visitor["Purchase_rate"] = (
    purchase_rate_by_visitor["Purchase_rate"] * 100
).round(2)

purchase_rate_by_visitor
```

VisitorType	Sessions	Purchases	Purchase_rate
Returning_Visitor	10551	1470	13.93
New_Visitor	1694	422	24.91
Other	85	16	18.82

Source: Author's analysis based on Sakar and Kastro (2018).

Description: Purchase rates varied by visitor type. New visitors recorded the highest purchasing rate, while returning users accounted for the majority of sessions and purchases.

Figure 9b. Purchase rates by month



```python
purchase_rate_by_month = (
    df_raw
    .groupby("Month", observed=True)
    .agg(
        Sessions=("Revenue", "size"),
        Purchases=("Revenue", "sum"),
        Purchase_rate=("Revenue", "mean")
    )
    .reindex(month_order)
)

purchase_rate_by_month["Purchase_rate"] = (
    purchase_rate_by_month["Purchase_rate"] * 100
).round(2)

purchase_rate_by_month
```

Month	Sessions	Purchases	Purchase_rate
Feb	184	3	1.63
Mar	1907	192	10.07
May	3364	365	10.85
June	288	29	10.07
Jul	432	66	15.28
Aug	433	76	17.55
Sep	448	86	19.20
Oct	549	115	20.95
Nov	2998	760	25.35
Dec	1727	216	12.51

Source: Author's analysis based on Sakar and Kastro (2018).

Description: Purchase rates also varied from month to month, with the purchase rate being highest in November. Purchases may be affected by seasonal factors; however, a descriptive comparison does not show a definite causal relation.

2.9 Data Understanding Summary

The findings related to the data source were positive and the dataset is usable for an example of a supervised classification method. It has the expected number of 12,330 sessions; no missing values; behavioural, technical and contextual variables are all included; exact matches were decided to be kept since, without session identifiers, there was no way to indicate that they were actually erroneous duplicates.

The target is imbalanced. Only 15.47% are purchases, while 84.53% are not. Integer-coded categories must be treated as categorical. Numerical measures would not need to be standardised. Initial comparison between different variables also suggests that PageValues is associated with the recorded Revenue outcome. Thus, the model is to be checked for predictive performance and at what level it relies on this variable.

3. Data Preparation

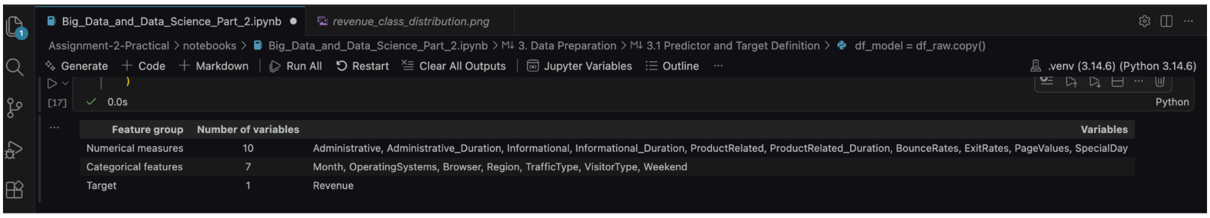
3.1 Predictor and Target Definition

In the next step, the data was split into the predictor matrix and target vector to get ready for model building. The binary target Revenue was turned into numbers, with 1 being ‘purchase’ and 0 being ‘no purchase’, to pass through some classification and evaluation functions later.

The 17 predictors were categorized not by their original data types, but according to the meaning of the measurement. Ten variables were treated as numerical measures: the page-count variables, their corresponding duration variables, BounceRates, ExitRates, PageValues and SpecialDay. The seven variables that represent categorical features are: Month, OperatingSystems, Browser, Region, TrafficType, VisitorType and Weekend.

This separation was needed as some of the categorical variables are stored using integers; viewing these numbers as continuous values would create an unnecessary ordering, whereas categorising them would support one-hot encoding down the line in the preprocessing stage, whilst allowing the numerical measures to be passed to the model as is without standardisation.

Figure 10a. Classification of predictor and target variables

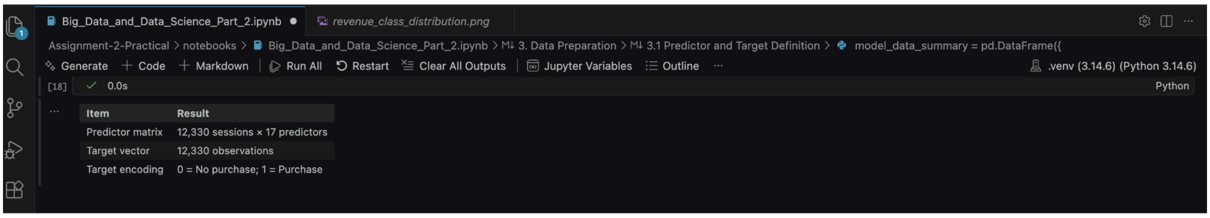


Feature group	Number of variables	Variables
Numerical measures	10	Administrative, Administrative_Duration, Informational, Informational_Duration, ProductRelated, ProductRelated_Duration, BounceRates, ExitRates, PageValues, SpecialDay
Categorical features	7	Month, OperatingSystems, Browser, Region, TrafficType, VisitorType, Weekend
Target	1	Revenue

Source: Author's analysis based on Sakar and Kastro (2018).

Description: In the output, each variable used for modelling purposes is categorised into its respective group according to its role in the analysis. The dataset has altogether ten numerical measures, seven categorical attributes and the Revenue target. This depicts that, before model training, the numerical attributes do not need categorical encoding, which needs to be performed for the categorical set of attributes.

Figure 10b. Predictor matrix and target encoding



Item	Result
Predictor matrix	12,330 sessions × 17 predictors
Target vector	12,330 observations
Target encoding	0 = No purchase; 1 = Purchase

Source: Author's analysis based on Sakar and Kastro (2018).

Description: The predictor matrix had 12,330 sessions and 17 predictors. The target vector had the 12,330 results indicating whether something was purchased or not. Revenue was set to 0 for no purchase and 1 for purchase.

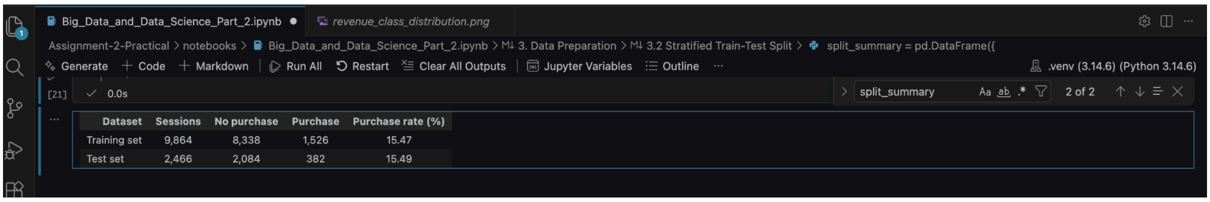
3.2 Stratified Train-Test Split

Data sets were split into training and test datasets prior to any preprocessing transformation being fitted. Preprocessing, parameter selection and model development have been done using the training data only; all test observations were separated and used only to evaluate the final model. Separating the testing observations minimizes the risk that information from the final testing dataset affects the model selection.

The data was partitioned into an 80:20 split, resulting in a training set of 9,864 sessions and a test set of 2,466 sessions. Stratification was implemented using the target variable to ensure a consistent class distribution in both splits. The train set consists of 1,526 purchases and the test set has 382. Purchasing sessions represent 15.47% of the training set and 15.49% of the test set. This is proportionate to the original dataset.

Additionally, the same training and test observations were retained for the later sensitivity analysis to compare models trained with and without PageValues using equal evaluation data. A fixed random state was used to obtain a reproducible split.

Figure 11. Stratified training and test datasets



Dataset	Sessions	No purchase	Purchase	Purchase rate (%)
Training set	9,864	8,338	1,526	15.47
Test set	2,466	2,084	382	15.49

Source: Author's analysis based on Sakar and Kastro (2018).

Description: The number of purchasing and non-purchasing sessions for the train set and test set, as allocated by the program, can be seen in the output. The stratification made the purchase rate of the training set roughly similar to the test set: 15.47% for the train set and 15.49% for the test set.

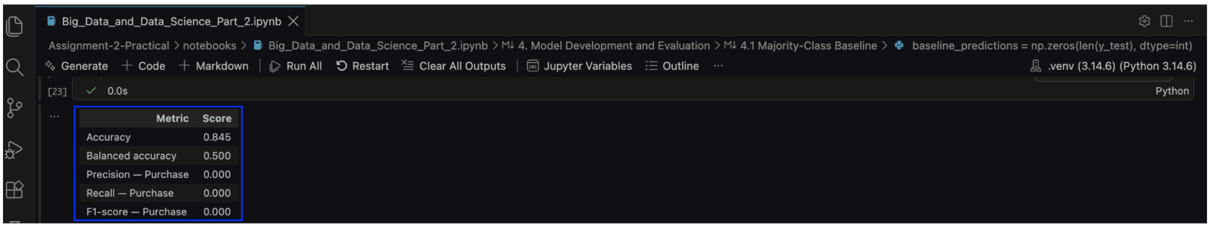
4. Model Development and Evaluation

4.1 Majority-Class Baseline

A Majority-Class Baseline was done before the step of training the Decision Tree. As most of the sessions did not result in a purchase, the majority-class baseline predicted for every observation in the test set that there would be 'No purchase'. This was a way of establishing a minimum against which the output of the predictive model could be compared.

The result was an accuracy of 84.5%, whereas the balanced accuracy was 50.0%. Similarly, precision, recall and F1-score for the purchase class were 0. Given these results, although the accuracy was high, the baseline failed to correctly classify any of the purchasing sessions from the test set, which consists of 382 purchasing sessions. This result verifies the previous statement that accuracy alone is not enough as an evaluation measure. This is caused by the imbalance of the problem.

Figure 12. Majority-class baseline performance



Metric	Score
Accuracy	0.845
Balanced accuracy	0.500
Precision — Purchase	0.000
Recall — Purchase	0.000
F1-score — Purchase	0.000

Source: Author's analysis based on Sakar and Kastro (2018).

Description: The baseline gives relatively high overall accuracy because it assigns every session to the majority class. But its 0 recall and F1-score for purchasing sessions show why balanced accuracy and purchase-class metrics are needed.

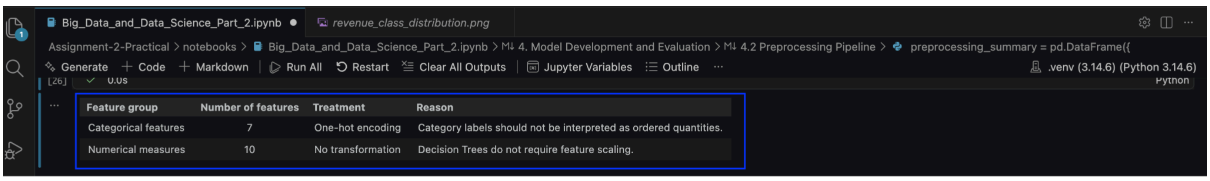
4.2 Preprocessing Pipeline

The scikit-learn pipeline incorporated preprocessing. One-hot encoding was used for transforming all seven categorical features, and ten numerical features were passed as they were. Previously unseen categories were ignored by the encoder. Thus, the pipeline does not fail with new observations containing an unseen category.

No imputation of missing values was needed, as there were no missing values in the source data. Numerical standardisation of the numerical predictors was found unnecessary. Decision Trees make classification decisions based on thresholds for certain variables and do not depend on distances between observations.

All preprocessing transformations were fitted on the training data only through the modelling pipeline. This prevented information from the test set influencing model development.

Figure 13. Preprocessing strategy for predictor groups



Feature group	Number of features	Treatment	Reason
Categorical features	7	One-hot encoding	Category labels should not be interpreted as ordered quantities.
Numerical measures	10	No transformation	Decision Trees do not require feature scaling.

Source: Author's analysis.

Description: There were seven categorical predictors, which were one-hot encoded, and ten numerical measures, which were left as they were. The transformations were put in the pipeline to make sure they were learnt only from the training data.

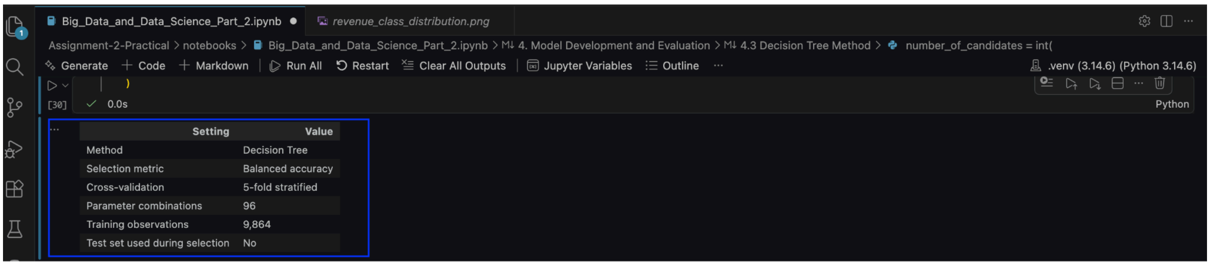
4.3 Decision Tree Method and Model Selection

The Decision Tree classifier was chosen as it can identify non-linear relationships and interactions among variables but keep the rules simple and therefore interpretable. At each interior node, it splits observations according to one predictor and a threshold that improve the separation of the classes.

Model selection was conducted using a grid search with stratified five-fold cross-validation. The search used two different splitting criteria, six different maximum-depth alternatives, four different minimum leaf sizes and two different class-weighting strategies, producing 96 parameter combinations (2 × 6 × 4 × 2 = 96). Balanced accuracy was used as the validation metric, as the focus is the ability of the model to predict both those who will purchase and those who will not purchase.

The cross-validation was applied only to the training-set part. The test set kept its initial role for testing after model tuning. It remained an independent verification set for the trained model.

Figure 14a. Decision Tree model-selection configuration



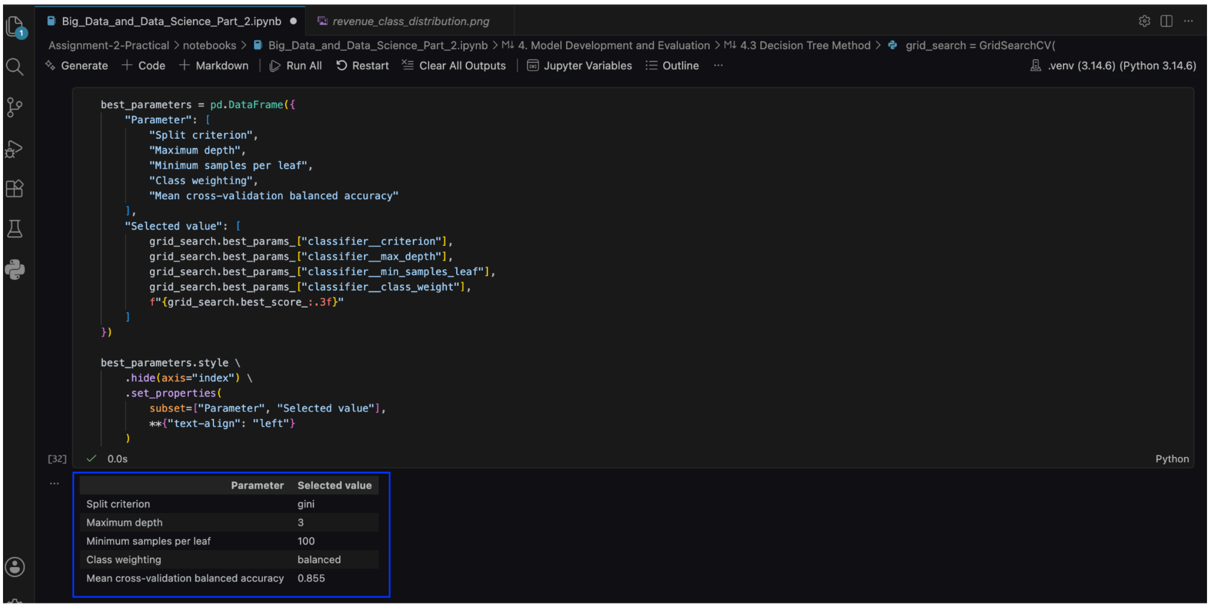
Setting	Value
Method	Decision Tree
Selection metric	Balanced accuracy
Cross-validation	5-fold stratified
Parameter combinations	96
Training observations	9,864
Test set used during selection	No

Source: Author's analysis.

Description: The configuration is a summary of the 96 parameter combinations evaluated using stratified five-fold cross-validation. For selecting a model capable of classifying and distinguishing between both classes, balanced accuracy was used.

The selected model used the Gini splitting criterion, a maximum depth of three, a minimum of 100 samples per leaf and balanced class weights. The balanced class weights gave more significance to the less common class of sessions that actually completed a purchase. The configuration achieved a mean cross-validated balanced accuracy of just under 85.5%.

Figure 14b. Parameters selected through cross-validation



Source: Author's analysis.

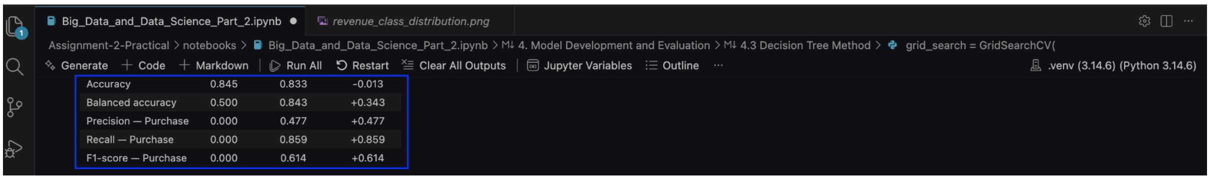
Description: The Gini criterion was used for the depth-limited and class-weighted Decision Tree. The restricted depth and minimum leaf size prevent an overly complex tree from being created, whereas balanced class weights attempt to account for the unequal distribution of outcomes.

4.4 Test-Set Performance

The trained Decision Tree pipeline was tested once on the independent test set after model selection. The Decision Tree attained overall accuracy equal to 83.3% and balanced accuracy of 84.3%, with 47.7% precision, 85.9% recall and an F1-score of 61.4% for the purchase class.

Relative to the majority-class baseline, the model produced a small decrease of about 1.2 percentage points in total accuracy, but a large increase of 34.3 percentage points in balanced accuracy. At the same time, purchase recall, which was zero for the baseline, increased to 85.9%. The loss of overall accuracy was the cost of the increase in the balance of the model, which was more willing to classify sessions as purchases than simply defaulting to the majority class.

Figure 15. Baseline and Decision Tree performance on the test set



Accuracy	0.845	0.833	-0.013
Balanced accuracy	0.500	0.843	+0.343
Precision — Purchase	0.000	0.477	+0.477
Recall — Purchase	0.000	0.859	+0.859
F1-score — Purchase	0.000	0.614	+0.614

Source: Author's analysis based on Sakar and Kastro (2018).

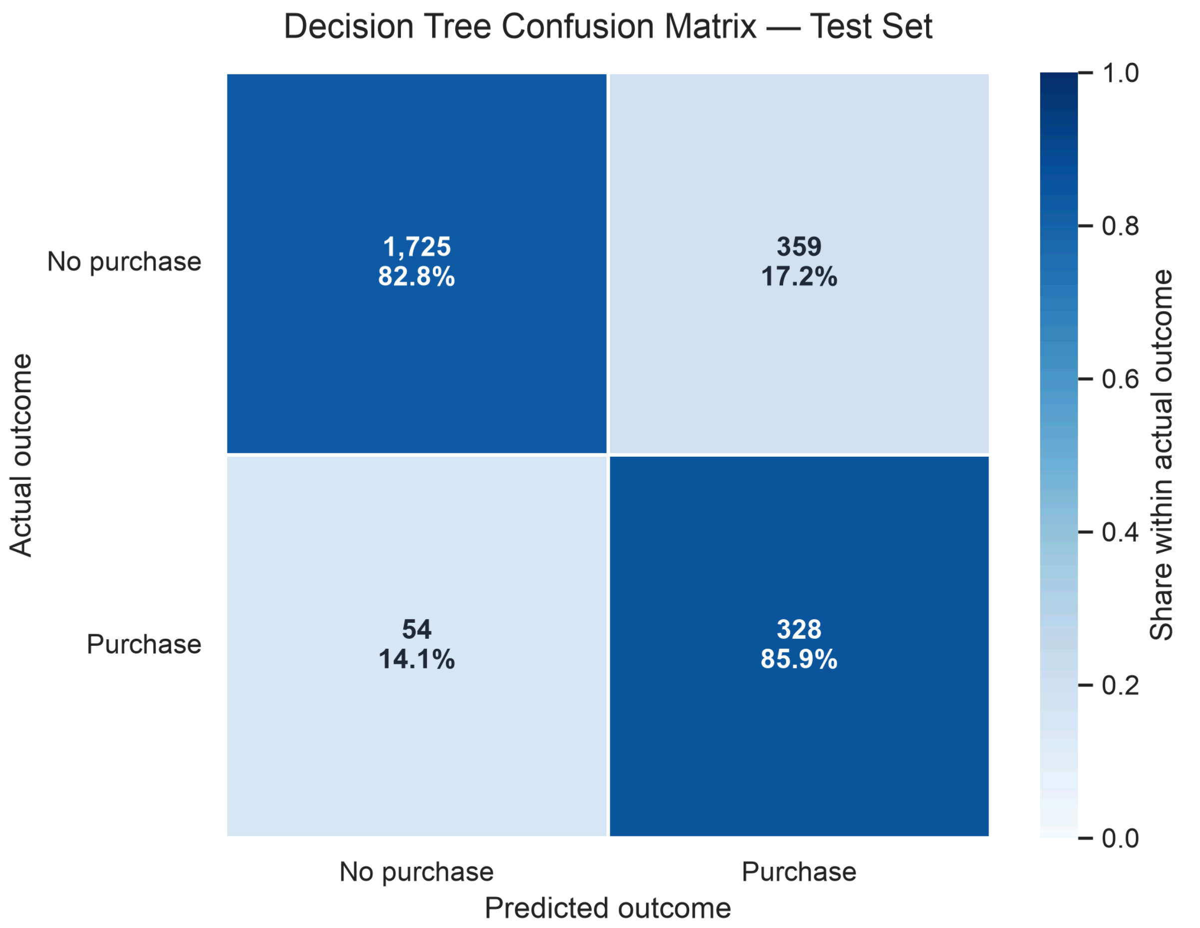
Description: The Decision Tree shows only slightly less total accuracy than the majority baseline classifier in exchange for superior balanced accuracy and improved results for all purchase-class metrics. This shows that the Decision Tree model has provided classification capability for the minority outcome.

4.5 Confusion Matrix

The confusion matrix provides more clarity about the model's predictions. From the 2,084 non-purchasing sessions in the test set, 1,725 were correctly labelled and 359 were incorrectly labelled as purchases. Furthermore, of the 382 purchasing sessions, 328 were spotted correctly and 54 remained undetected.

These results give a purchase recall of 85.9% and a non-purchase recall of approximately 82.8%. The model identified both states in a relatively even balance, although its sensitivity to purchases came with a meaningful number of false-positive purchase predictions.

Figure 16. Decision Tree confusion matrix on the test set



Source: Author's analysis based on Sakar and Kastro (2018).

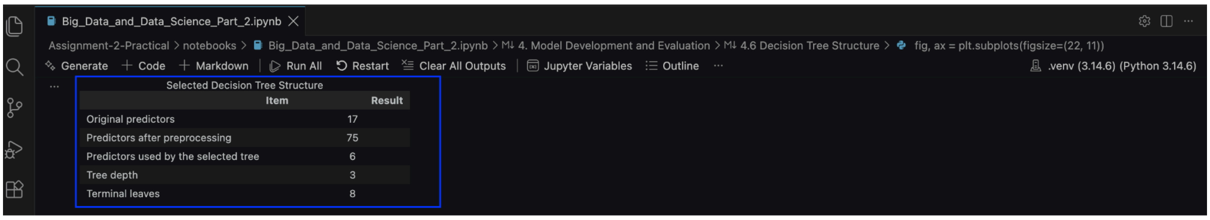
Description: The model correctly predicts 328 out of 382 instances when a purchase occurred and 1,725 out of 2,084 when a purchase did not occur. Furthermore, it exhibits the trade-off between identifying purchases and generating 359 incorrectly predicted purchases.

4.6 Decision Tree Structure

The number of model inputs expanded to 75 from the original 17 due to one-hot encoding, although only six were chosen for use as splits in the selected tree. The resulting model had a depth of three and eight terminal leaves, which produced a relatively compact Decision Tree.

At the root of the tree, PageValues was placed as the first variable used in splitting the outcomes. Other variables that could be seen in the tree were ProductRelated_Duration, BounceRates and indicators for the months of May, September and November.

Figure 17a. Structural summary of the selected Decision Tree



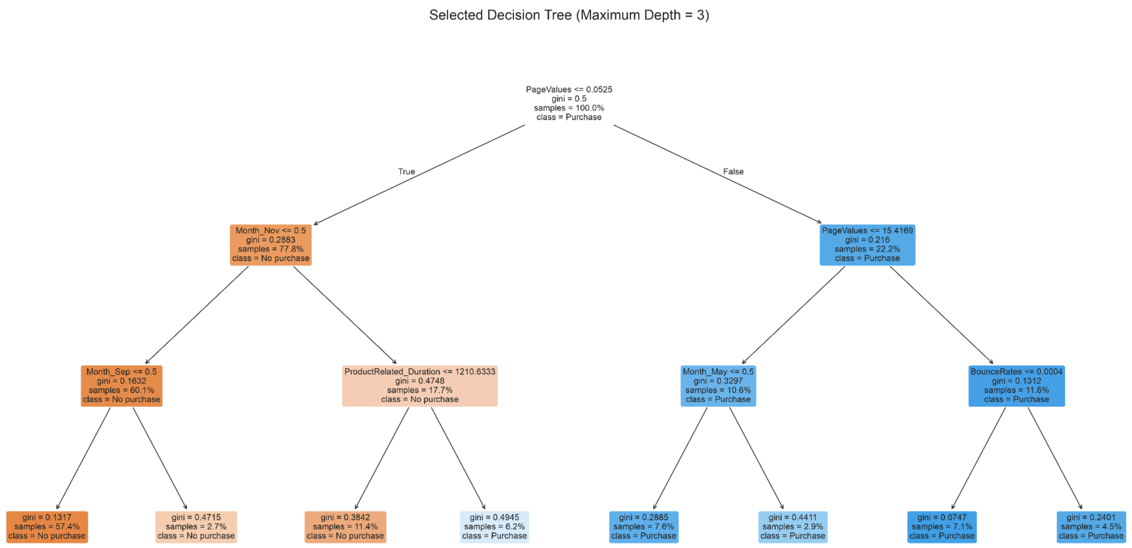
Selected Decision Tree Structure

Item	Result
Original predictors	17
Predictors after preprocessing	75
Predictors used by the selected tree	6
Tree depth	3
Terminal leaves	8

Source: Author's analysis.

Description: The processing stage uses six of the total 75 encoded inputs. Preprocessing converted the 17 original predictors into 75 inputs. The tree has a depth of three and eight terminal leaves, resulting in a relatively low number of rules.

Figure 17b. Selected Decision Tree structure



Source: Author's analysis based on Sakar and Kastro (2018).

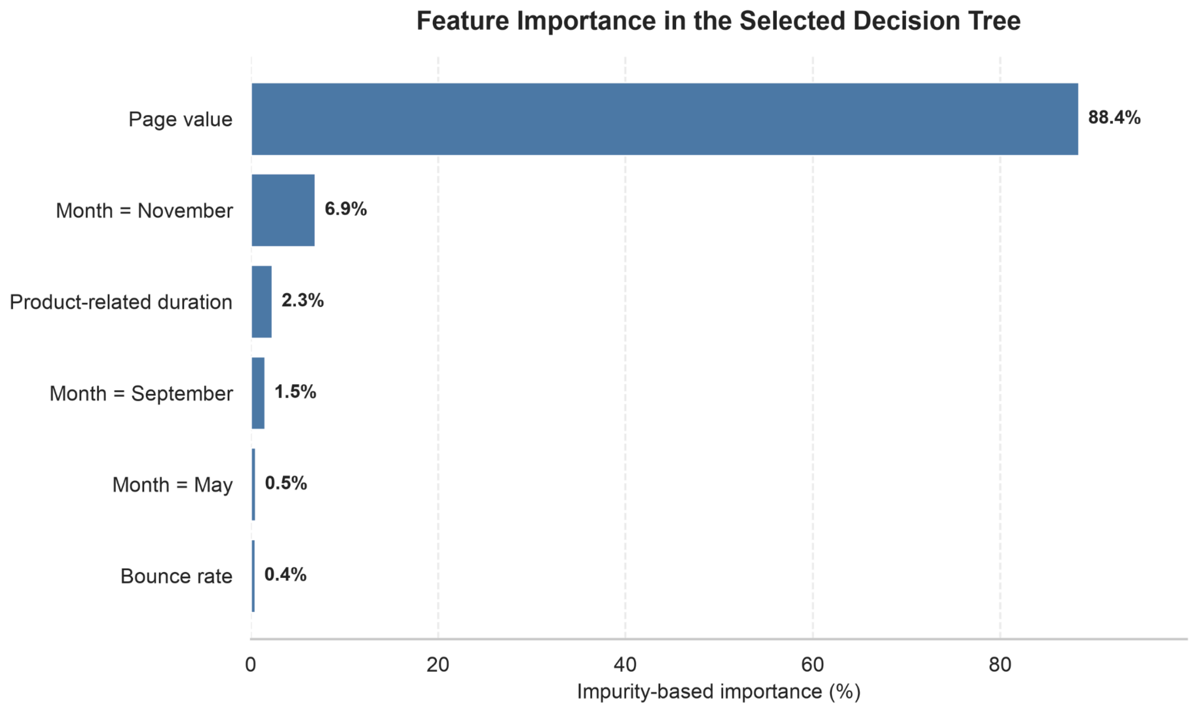
Description: The tree visualises the sequence of rules used to classify the sessions. The initial rule that forms the root node is associated with PageValues and is followed by decisions on session duration, bounce behaviour and chosen monthly indicators.

4.7 Feature Importance

The feature-importance results reiterate the importance of PageValues for the model. It came up as having around 88.4% of the total impurity-based importance. The second-largest contributor, the November indicator, had around 6.9%. ProductRelated_Duration, September, May and BounceRates made minor contributions.

These importance values indicate the extent to which the fitted tree used each variable to build splits. This is not to be interpreted as evidence that a predictor causes a purchase.

Figure 18. Feature importance in the selected Decision Tree



Source: Author's analysis based on Sakar and Kastro (2018).

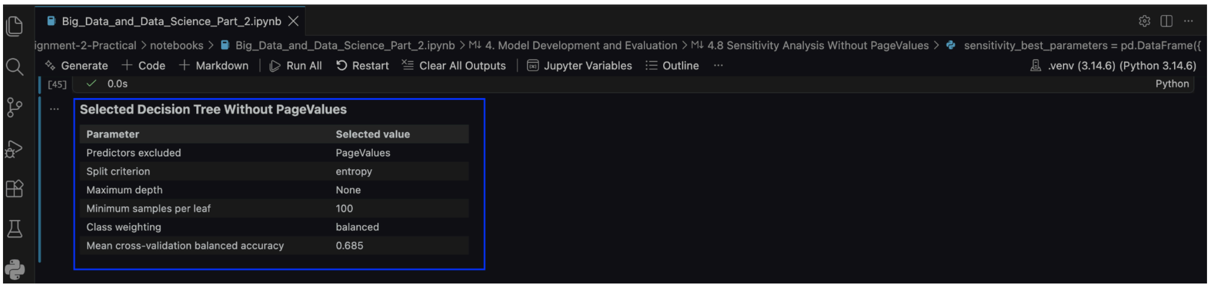
Description: PageValues dominates impurity-based importance, whereas the remaining variables are less important. The values explain the use of predictors by the model without establishing causality between them.

4.8 Sensitivity Analysis Without PageValues

As PageValues appeared to dominate the tree structure and feature-importance outcomes, a sensitivity analysis excluding it sought to test the model's reliance on this predictor. All models were retrained on identical training and test observations with an identical cross-validation design, parameter grid and evaluation measures.

In the absence of PageValues, cross-validation found an entropy-based tree with balanced class weights, a minimum leaf size of 100 and an unconstrained maximum depth to have the highest performance. This model achieved a mean cross-validated balanced accuracy of approximately 68.5%.

Figure 19a. Selected Decision Tree configuration without PageValues


Selected Decision Tree Without PageValues

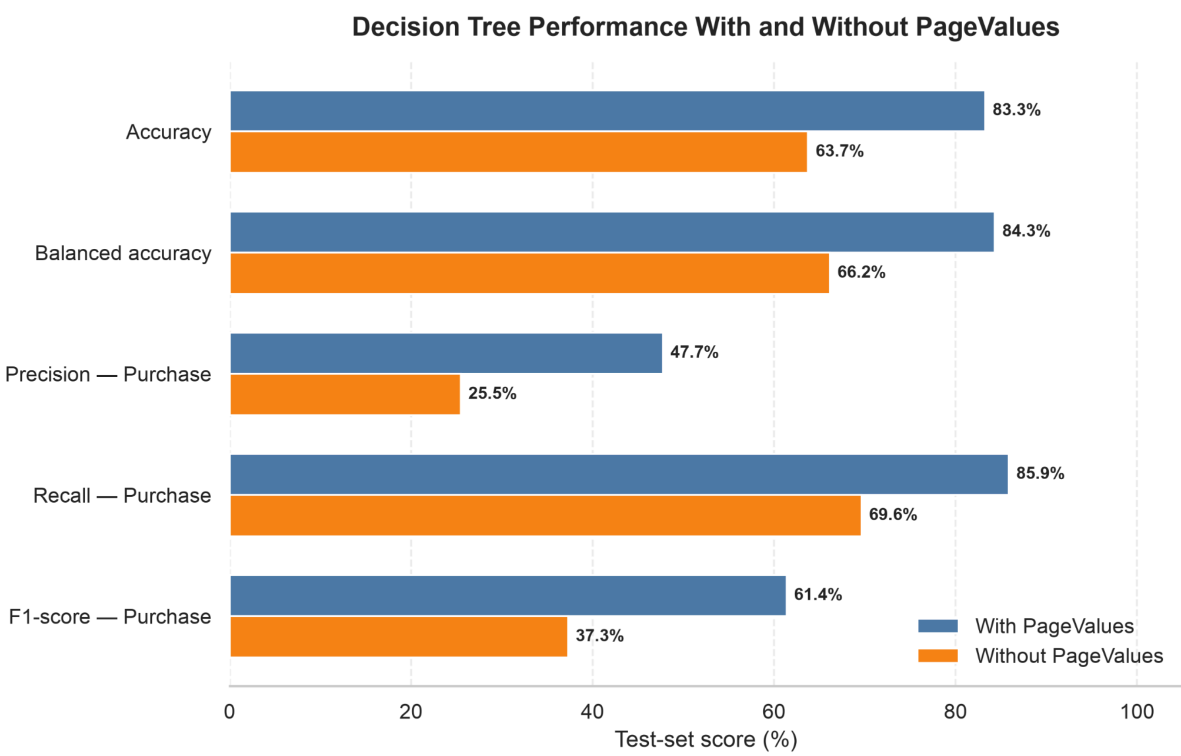
Parameter	Selected value
Predictors excluded	PageValues
Split criterion	entropy
Maximum depth	None
Minimum samples per leaf	100
Class weighting	balanced
Mean cross-validation balanced accuracy	0.685

Source: Author's analysis.

Description: Removing PageValues generates a different and less effective model configuration. The lower cross-validated balanced accuracy indicates that the remaining session characteristics have much less predictive power.

On the test set, balanced accuracy decreased from 84.3% to 66.2%, and the purchase-class F1-score decreased from 61.4% to 37.3%. Purchase recall decreased from 85.9% to 69.6%, while purchase precision decreased from 47.7% to 25.5%.

Figure 19b. Model performance with and without PageValues

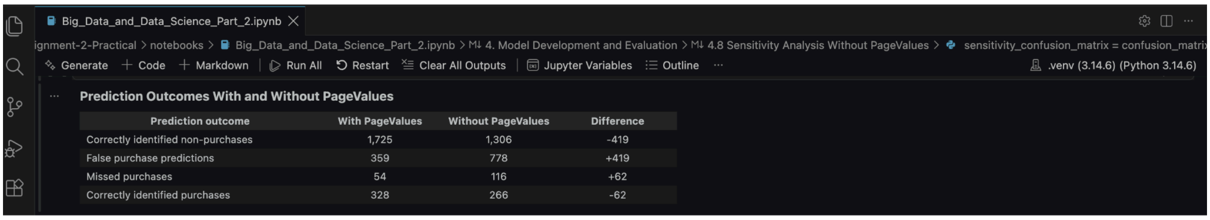


Source: Author's analysis based on Sakar and Kastro (2018).

Description: Balanced accuracy and, in particular, the purchase-class F1-score show drops that reveal the model's deep reliance on the values stored in this field.

The number of correctly predicted purchasing sessions decreased from 328 to 266, whereas missed purchases rose from 54 to 116. Furthermore, false-positive predictions of purchase grew from 359 to 778. The sensitivity analysis suggests that the predictor plays a role in detecting both outcomes.

Figure 19c. Prediction outcomes with and without PageValues



Prediction Outcomes With and Without PageValues

Prediction outcome	With PageValues	Without PageValues	Difference
Correctly identified non-purchases	1,725	1,306	-419
False purchase predictions	359	778	+419
Missed purchases	54	116	+62
Correctly identified purchases	328	266	-62

Source: Author's analysis based on Sakar and Kastro (2018).

Description: Removing PageValues decreases the number of correct classifications for both outcomes and increases the number of both false-positive and false-negative predictions. The comparison quantifies the practical effect of the degradation in performance reported in Figure 19b.

5. Discussion of Results

Comparing the above with the base majority-class classifier, it can be seen precisely why accuracy is a poor measure of performance for this type of problem, as while accuracy was 84.5%, the classifier failed completely to predict purchase of any session at all, while the chosen decision tree albeit with a slightly lower accuracy of 83.3% raised balanced accuracy from 50.0% to 84.3%, predicting 328 of the 382 purchases correctly.

The practical trade-off is this: purchase recall shoots to 85.9%; however, 359 false purchase predictions resulted in precision of 47.7%, meaning barely 48 of the 100 sessions deemed as purchasing sessions by the model turned out in an actual sale. Although this performance can be perfectly acceptable for low cost action like recommendation or prompt of service, an expensive discount or incentive would not be as suitable. Preference should depend on cost of a missed purchase and irrelevant discount provided, so balance between recall and precision can be biased one way or another.

With maximum depth restricted to 3 and minimum leaf size to 100, the resulting tree is small and more interpretable. This helps restrict the complexity of model and might mean that more subtle behavioural interactions are not modelled. Stratified cross validation was used for model selection and the independent test set provides an extra set of performance figures.

Model performance primarily depended on PageValues. Balanced accuracy reduced from 84.3% to 66.2% when the feature was omitted, while purchase F1-score reduced from 61.4% to 37.3%. False and missed purchase predictions increased from 359 to 778 and 54 to 116, respectively. The remaining behavioural and contextual predictors carried useful, yet not equally discriminative information.

This reliance comes at a functional cost. If the model is to assist an early intervention, PageValues must be computed before that intervention occurs, as otherwise the performance of the full model would not be valid for an early-prediction task. The model therefore is left with no option other than trade-off its predictive performance for the features to be available to it, as the conclusion from the sensitivity analyses suggests.

In addition, there were also source data limitations which were out of human control - without session id, there was no way to identify multiple records with same exact entries as erroneous duplications. Sessions were also rolled up meaning the models never obtained sequence or timing information. Moreover, insights obtained using one dataset during a specific period using a particular type of modeling method can never readily transfer to another retailer, period or customer population.

It could be beneficial in identifying potential purchases; but could also miss crucial last-minute changes to customers' likelihoods. Lastly, any practical application of this model depends on its business impact that combines with both costs of intervention (marketing or similar) and costs of incorrect classification outcomes.

6. Conclusion

This practical assignment used an interpretable Decision Tree model to predict purchases from e-commerce session data, understand the source data and prepare potential predictors, split the data using stratified sampling, cross-validate to select a suitable model and then obtain the evaluation metrics on an unseen test set.

The baseline was majority-class, which gave 84.5% accuracy but identified no sessions which actually came from a purchase. Conversely, the model chosen provided a balanced accuracy of 84.3%, identifying 85.9% of actual purchasing sessions. Therefore, these measures provide a more informative estimate of performance, as accuracy alone concealed the baseline's failure to identify the purchase class.

PageValues was the most predictive variable and the removal of PageValues affected the results substantially, so its usefulness might depend upon whether PageValues is calculated and available prior to the planned action. The developed model may support low-cost actions such as product recommendations or service prompts, but the costs of wrong predictions should be borne in mind before deployment.

Comparison of other classifiers, adjusting the decision threshold based on business costs and exploration of predictors, which would be available earlier in the session, may be explored. The model should ideally be validated based on more recent or organisation-specific data prior to deployment. Clickstream-related data, if evaluated using appropriate methods, can support useful purchase predictions when the intended business decision is also considered.

7. References

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

Sakar, C. O., Polat, S. O., Katircioglu, M., & Kastro, Y. (2019). Real-time prediction of online shoppers' purchasing intention using multilayer perceptron and LSTM recurrent neural networks. Neural Computing and Applications, 31(10), 6893–6908. https://doi.org/10.1007/s00521-018-3523-0

Sakar, C., & Kastro, Y. (2018). Online Shoppers Purchasing Intention Dataset [Data set]. UCI Machine Learning Repository. https://doi.org/10.24432/C5F88Q

8. List of Figures

Figure 19c. Prediction outcomes with and without PageValues